# Decentralized E-Voting System using Blockchain Technology

Group Members:
Abhinav Raj P V
Amith Murali
Arun P V
Niketh Vijay K
Muhammed C

Supervised by: Dr. Ajish Kumar K S
Department of Computer Science
Government College of Engineering Kannur

February 13, 2026

# Introduction

## Context

- Secure and transparent voting is vital in modern democracies.
- Traditional electronic voting systems are vulnerable to manipulation and lack of transparency.
- Integrating blockchain with biometric verification ensures secure and trusted voting.

## Why Blockchain + Face Recognition?

- Blockchain ensures immutability and transparency.
- Facial recognition guarantees one-person-one-vote.
- Dual-layered approach reduces fraudulent voting and increases public trust.

# Problem Statement

## Project Problem Statement

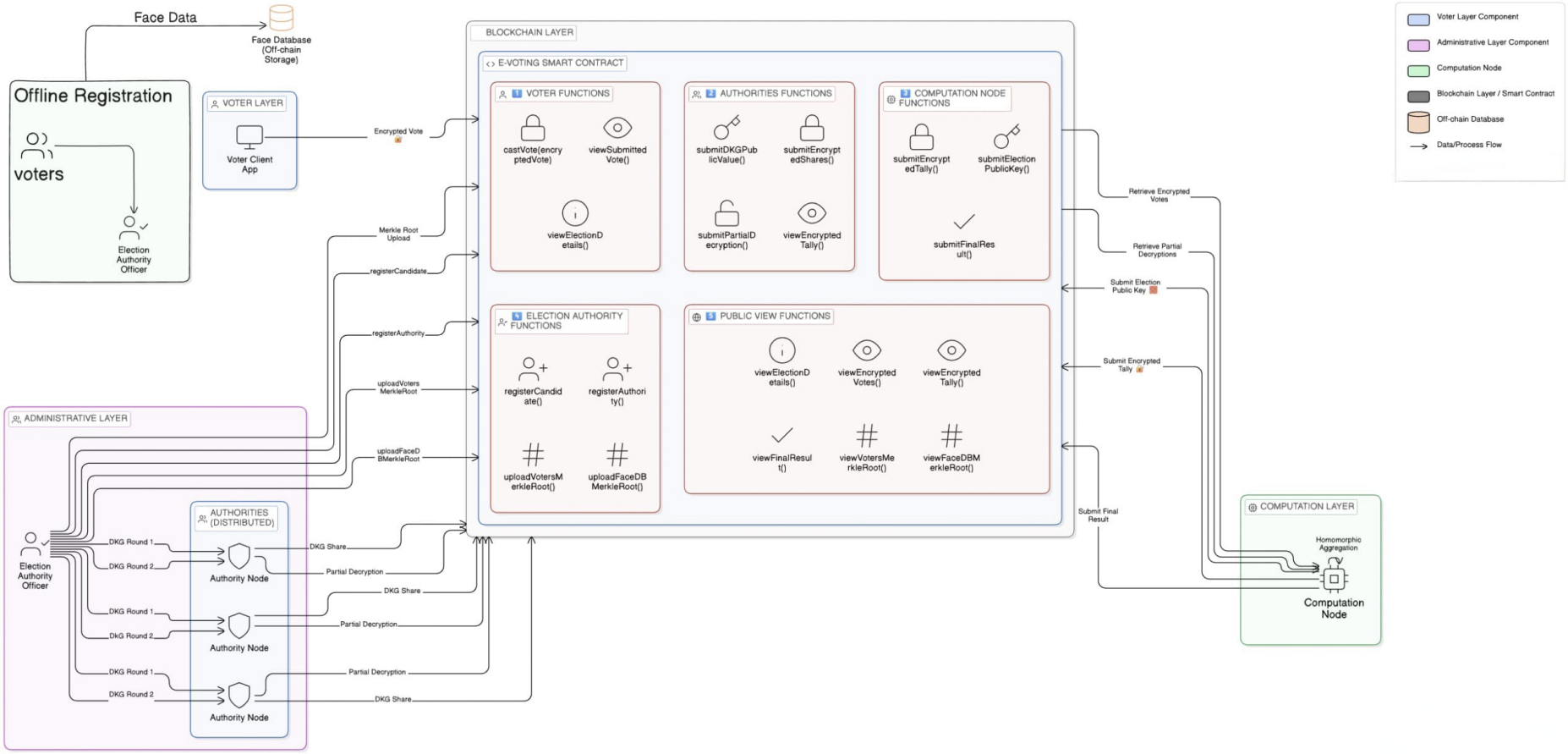
To develop a secure and transparent blockchain-based voting system that uses facial recognition for voter authentication and **Ethereum Blockchain** for immutable vote recording. The system aims to prevent fraud, duplication, and ensure a single verifiable vote per authenticated user.

# Project Objectives

## Key Goals of the Project

- To develop a Facial Recognition Module for voter identity verification.
- To implement **Ethereum Blockchain** for decentralized, tamper-proof vote storage.
- To design and deploy smart contracts for vote casting and validation.
- To securely store encrypted votes on the blockchain.
- To provide a voter dashboard to confirm vote status and transparency.

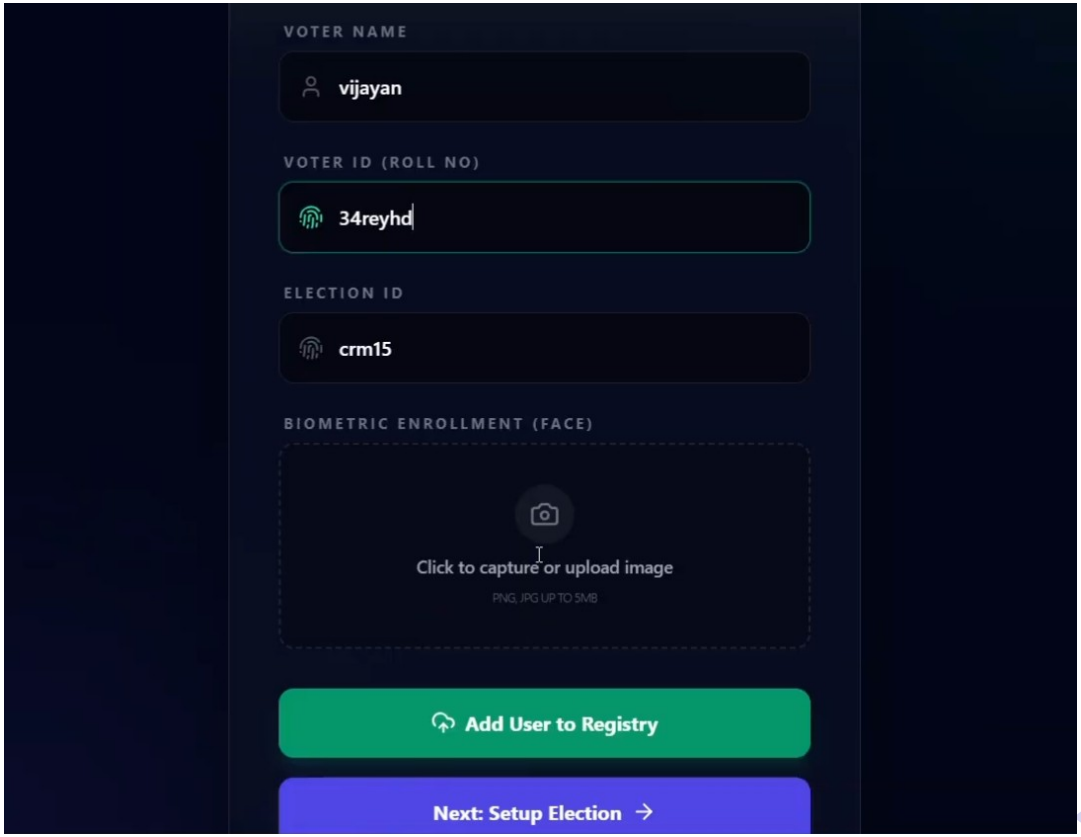
# Overall System Architecture



# Implementation: Admin Registers Voters

## Admin-Side Voter Registration (Offline Verification)

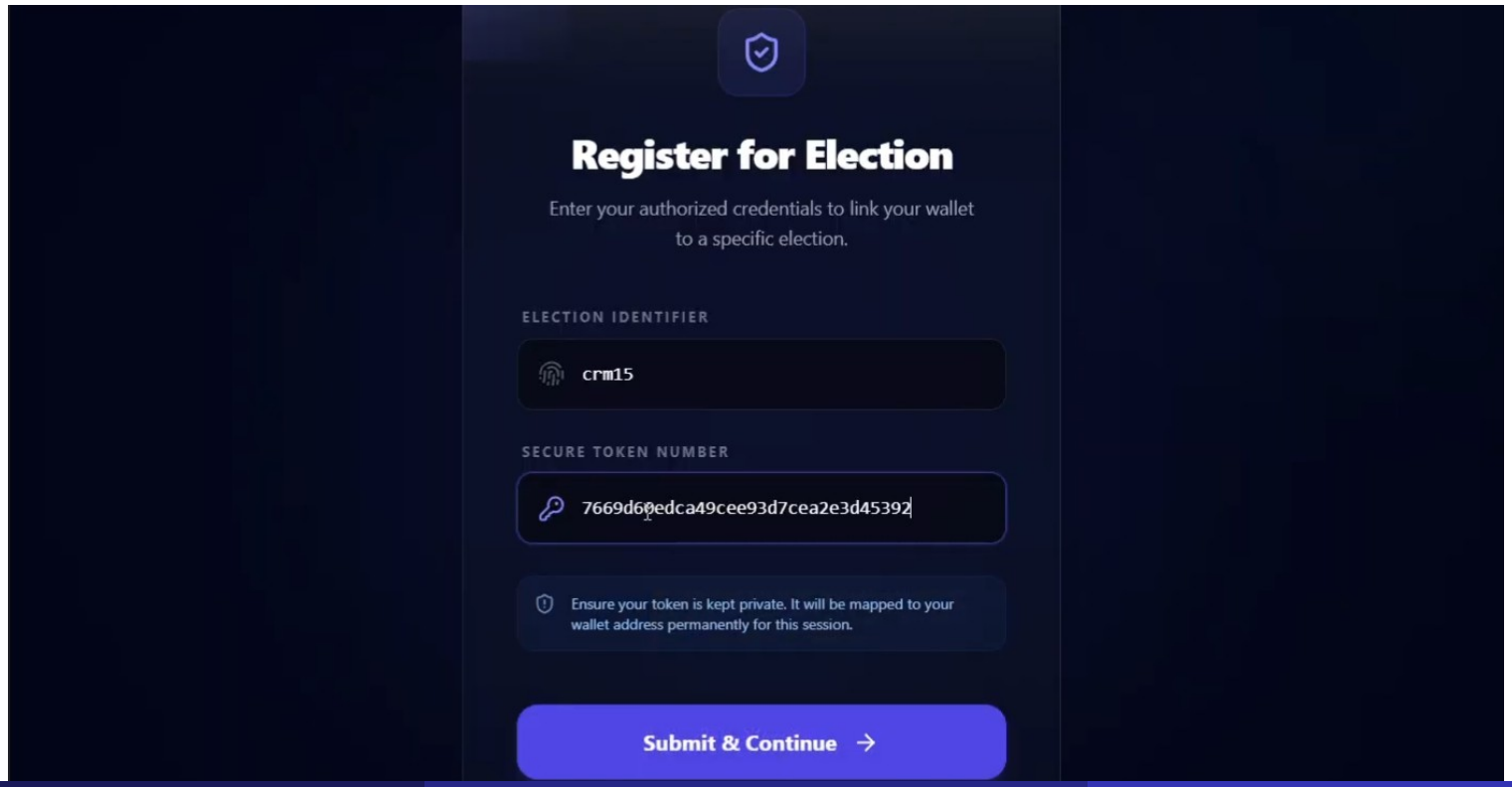
- Admin verifies voter identity physically (offline).
- Admin enters Voter Name,Voter ID and Election ID
- A unique secure token is generated for the voter.



# Implementation: Voter Secure Token Entry

## Token-Based Voter Authorization

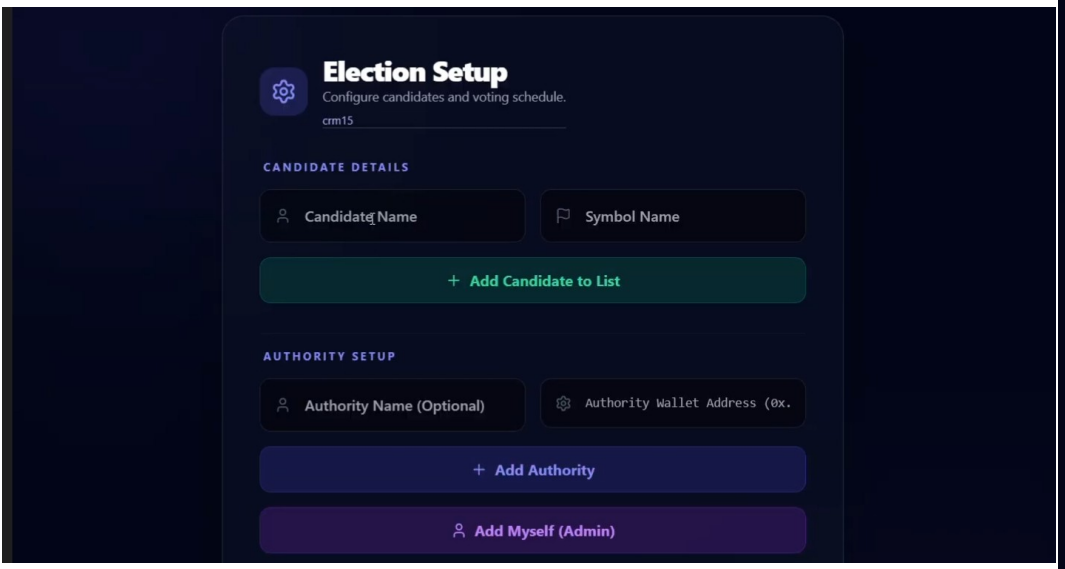
- Voter enters Election ID and the Secure Token Number(Generated at the time of Registration) into his personal system.
- Token verifies that the voter is authorized.
- Voter wallet is linked to the specific election.

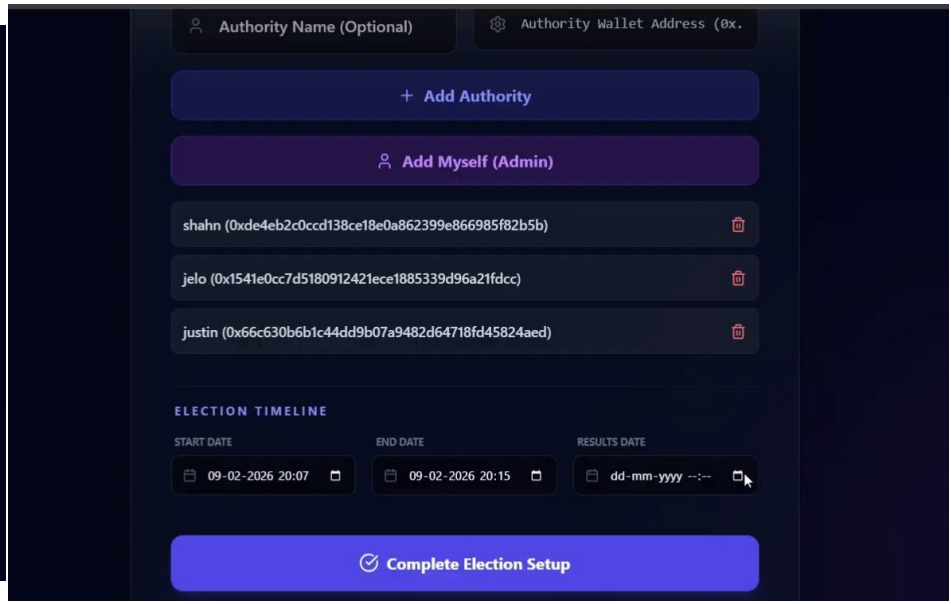


# Implementation: Election Setup by Admin

## Election Configuration Module

- Admin enters **Candidate Details(Name,Symbol Name)**
- Admin add **Election Authorities(Name,Wallet Address)**
- Admin defines **Election Timeline(Start Date,End Date ,Result Declaration Date)**

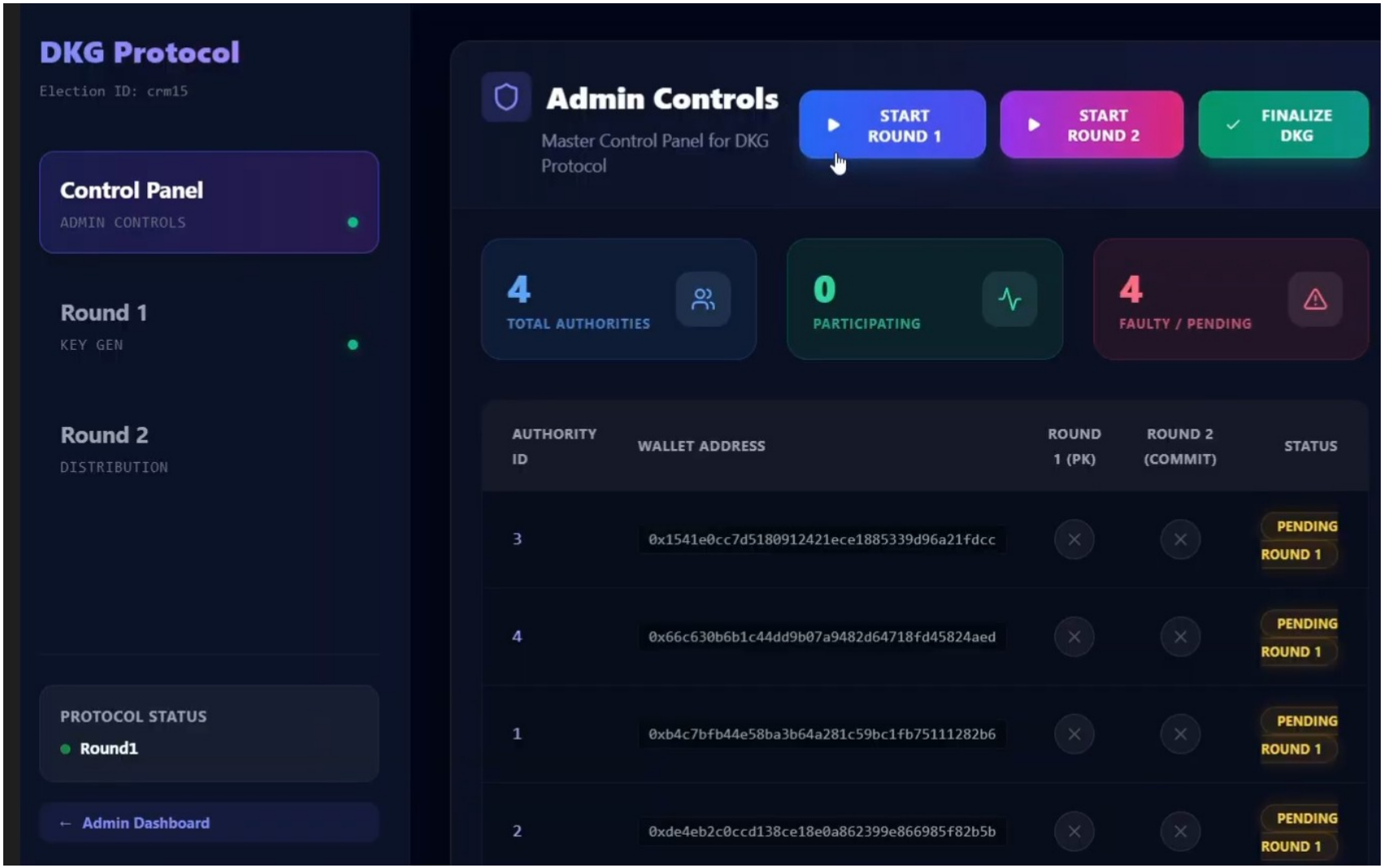


# Implementation: Distributed Key Generation

## DKG Initialization by Admin

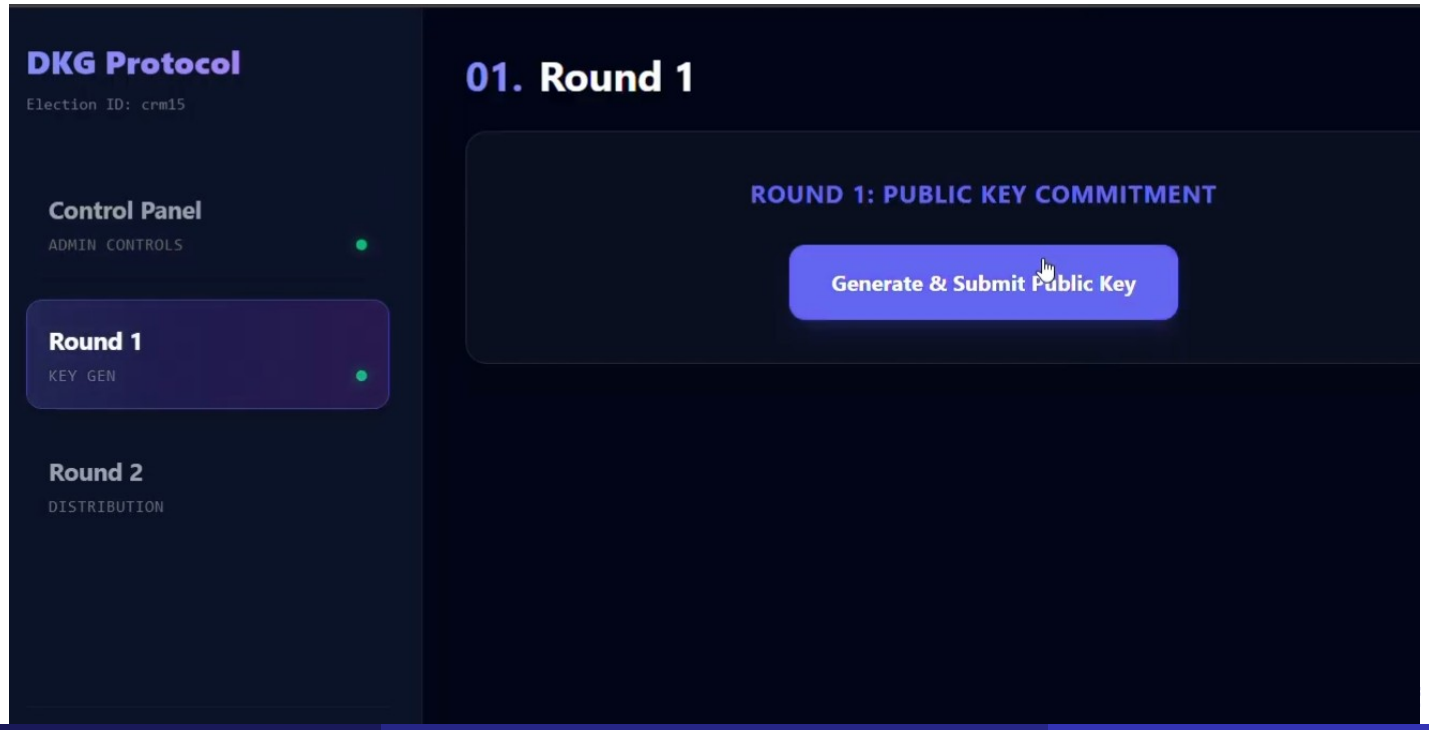
- Election requires a Public Key and Private Key for secure vote encryption and decryption.
- Instead of generating a single key centrally, a Distributed Key Generation (DKG) protocol is used.
- Admin initiates the DKG process through the control panel.
- Multiple authorities participate in key generation.


DKG Protocol
Election ID: crm15
Control Panel
ADMIN CONTROLS
Round 1
KEY GEN
Round 2
DISTRIBUTION
PROTOCOL STATUS
Round1
← Admin Dashboard
Admin Controls
Master Control Panel for DKG Protocol
START ROUND 1
START ROUND 2
FINALIZE DKG
4
TOTAL AUTHORITIES
0
PARTICIPATING
4
FAULTY / PENDING
AUTHORITY ID
WALLET ADDRESS
ROUND 1 (PK)
ROUND 2 (COMMIT)
STATUS
3
0x1541e0cc7d5180912421ece1885339d96a21fdcc
PENDING ROUND 1
4
0x66c630b6b1c44dd9b07a9482d64718fd45824aed
PENDING ROUND 1
1
0xb4c7bfb44e58ba3b64a281c59bc1fb75111282b6
PENDING ROUND 1
2
0xde4eb2c0ccd138ce18e0a862399e866985f82b5b
PENDING ROUND 1

# Implementation: DKG Round 1 – Public Key Commitment

## Round 1: Individual Key Generation

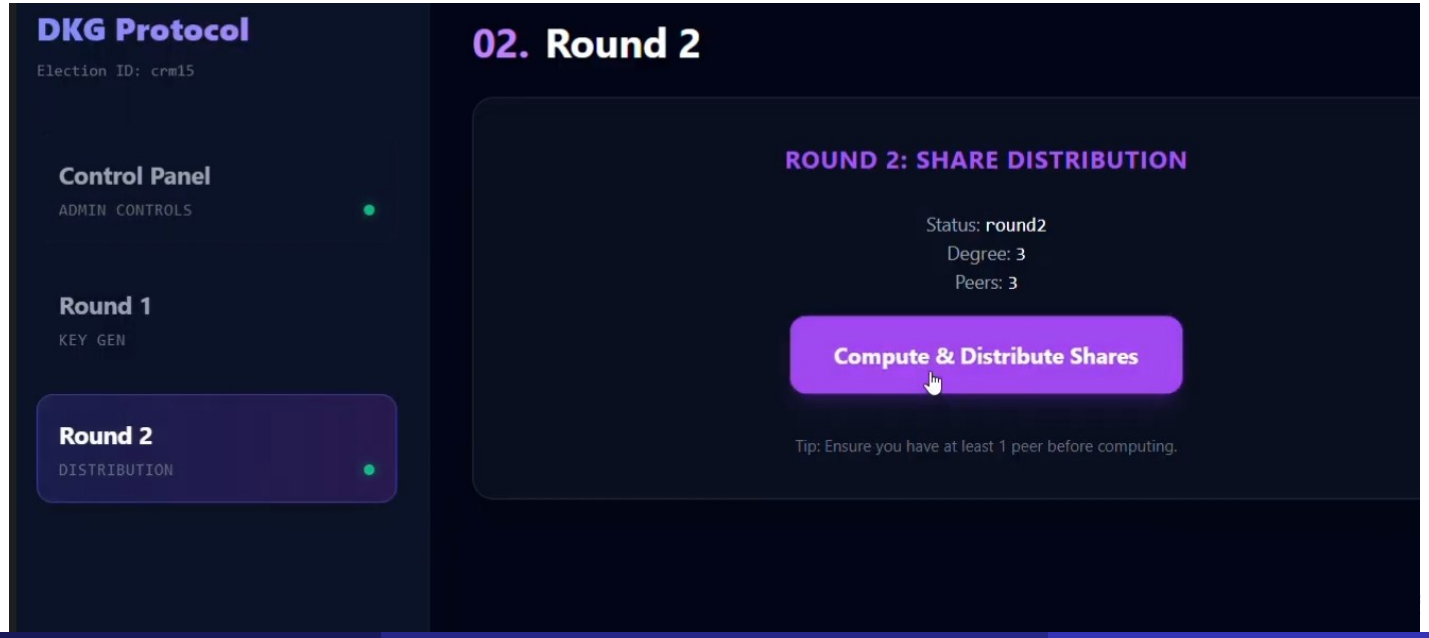
- Admin starts Round 1 of DKG.
- Each authority (including Admin):
  - Generates a public-private key pair.
  - Submits the public key.
- Private keys remain secret with respective authorities.



# Implementation: DKG Round 2 – Share Computation

## Secret Share Computation and Distribution

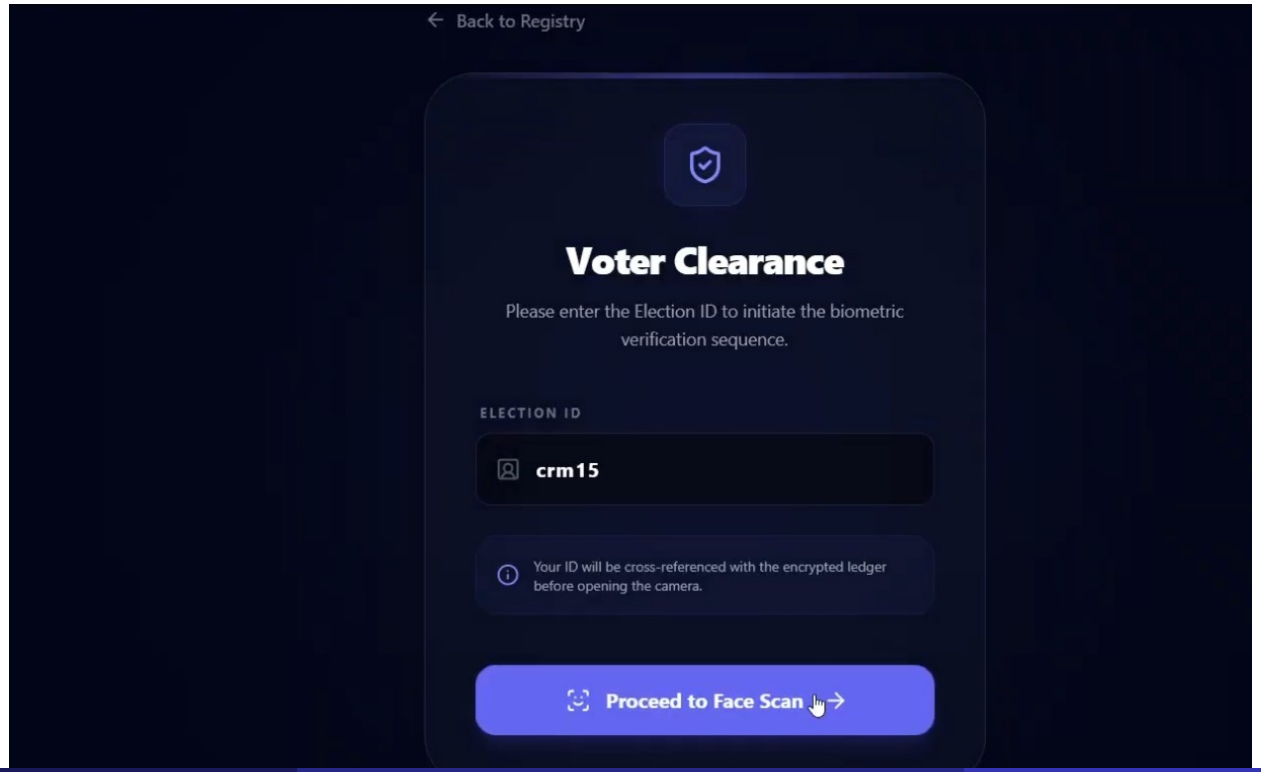
- In Round 2, each authority clicks **Compute & Distribute Shares**.
- Secret shares are computed for all participating authorities.
- Shares collectively generate:
  - Election Public Key
  - Distributed Election Private Key
- No single authority holds the complete private key.



# Implementation: Voter Election Entry and Verification

## Voter Access to Election

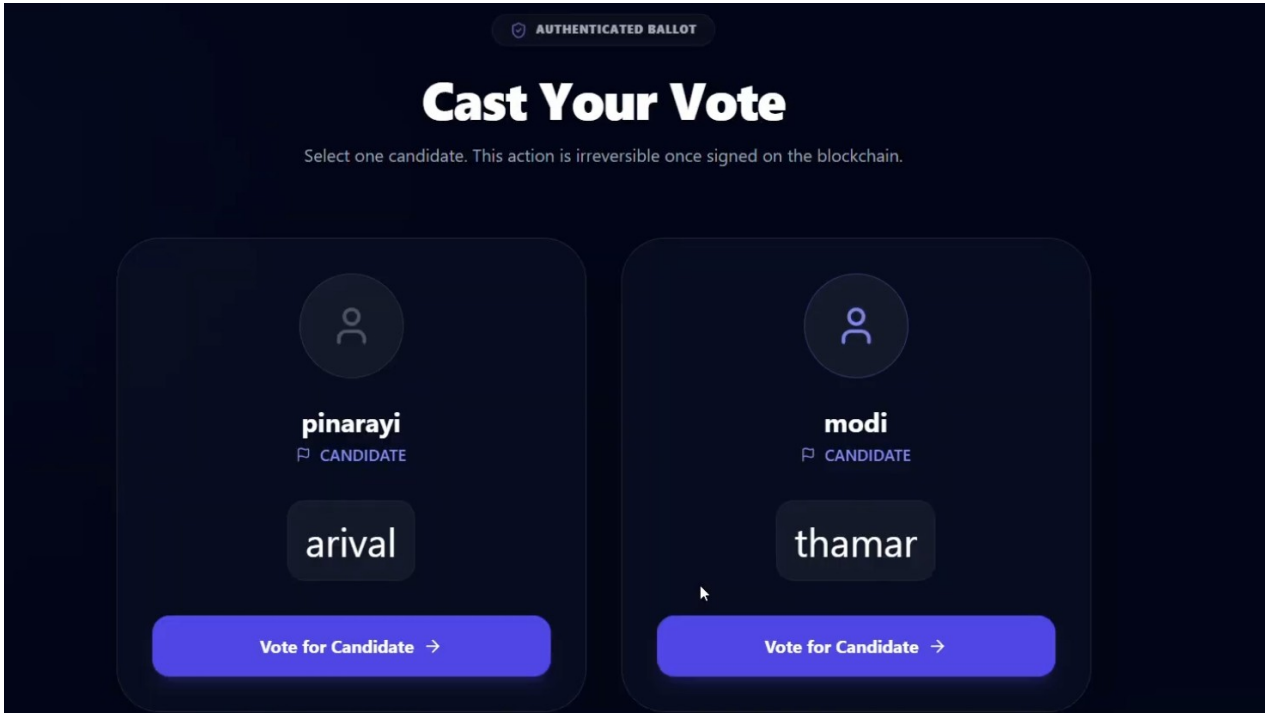
- After election setup and activation, the voter logs into the system.
- Voter selects the registered election and enters the Election ID.
- System initiates biometric verification step.
- Face scan integration is planned (currently under development).



# Implementation: Vote Casting and Encryption

## Secure Vote Casting Process

- After verification, voter is presented with the list of candidates.
- Voter selects a candidate and confirms the vote.
- The selected vote is encrypted using the Election Public Key.
- Encrypted vote is stored.



# Implementation: Result Decryption and Publishing

## Post-Election Result Workflow

- After election ends, Admin and authorities click **Calculate Result**.
- System aggregates all encrypted votes using homomorphic encryption.
- Decryption shares are submitted to reconstruct the final tally.
- Final vote count is displayed and published for public viewing.

# Project Objectives Completed So Far

## Objectives Successfully Achieved

- **Ethereum Blockchain Implementation**
  - Implemented decentralized, tamper-proof vote storage.
  - Configured blockchain environment for vote recording.
- **Smart Contract Design and Deployment**
  - Designed smart contracts for vote casting.
  - Implemented vote validation logic.
- **Voter Dashboard**
  - Developed dashboard to display vote status.
  - Enabled transparency for voters to verify their vote submission.

# Project Objectives Yet To Be Completed

## Remaining Objectives

- **Facial Recognition Module**
  - Implement biometric-based voter identity verification.
  - Integrate facial recognition with authentication workflow.
- **Storing Vote on Blockchain**
  - Implement secure recording of encrypted vote on blockchain.

# Conclusion

## Conclusion

- Successfully implemented the core modules of the decentralized e-voting system.
- Integrated Distributed Key Generation, vote encryption, homomorphic tallying, and threshold decryption.
- Enabled secure vote casting and transparent result publishing.
- Facial recognition integration and Blockchain Storage remain as future work.

# References

## Selected References


1. I. Stancíková & I. Homoliak, "SBvote: Scalable Self-Tallying Blockchain-Based Voting," Proceedings of the 38th ACM/SIGAPP Symposium on Applied Computing (SAC '23), March 2023.
2. J. Zhang, C. Wu, R. S. Sherratt, J. Wang, "An Improved Secure and Efficient E-Voting Scheme Based on Blockchain Systems," IEEE Internet of Things Journal, 2024.
3. Yadav, A.S. et al., "E-Voting using Blockchain Technology," International Journal of Engineering Research & Technology (IJERT), Vol. 9, No. 7, July 2020.